Week 5 – Quantum Computing

No-cloning, Deutsch Algorithm & The CHSH Game

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Review from Week 4

- Last week, we observed that a two-qubit system in the computational basis is of the form

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

where $a, b, c, d \in \mathbb{C}$ and $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$, and $|xy\rangle$ (or $|x\rangle|y\rangle$) is frequently used as shorthand for $|x\rangle \otimes |y\rangle$, for $x, y \in \{0, 1\}$.

- In the above system, the probability of measuring the state, say $|01\rangle$ (in the computational basis) is $|b|^2$; similarly, the probability of measuring 0 on qubit 1 (or wire 1) is $|a|^2 + |b|^2$, and the system collapses to:

$$\frac{a|00\rangle + b|01\rangle}{\sqrt{|a|^2 + |b|^2}}.$$

1. Consider the **product** state given by $|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{\sqrt{8}}|10\rangle + \sqrt{\frac{3}{8}}|11\rangle$. Note that the probability of measuring 0 in qubit 1 is $1/8 + 3/8 = 1/2$; similarly for measuring 1 in qubit 2.

   Now, if the second qubit is measured, and if the outcome observed (with probability 3/4) is 1, then the new state of the system is given by $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$. Note that the probability of measuring outcome i ($i \in \{0, 1\}$) in qubit 1 is still 1/2.

2. Consider now the **entangled** state given by $|\Phi^+\rangle = \dfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. If the outcome observed (as a result of measurement in the computational basis) in qubit 1 is 0, then the state collapses to $|00\rangle$.

Note: The phenomenon of entanglement will be key to understanding both the theory of quantum information and computing as well as its most powerful applications.

No-cloning theorem

Theorem 1. *There does not exist a unitary operator U such that for any qubit $|\psi\rangle$*

$$U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle.$$

We will see two proofs of this result – the first proof is simpler to grasp, but the second one is more conceptually insightful.

Proof # 1. By way of contradiction, suppose that there does exist a "universal cloner" U. In particular, we know that it must clone the two computational basis states, i.e. $U(|0\rangle|0\rangle) = |0\rangle|0\rangle$, and $U(|1\rangle|0\rangle) = |1\rangle|1\rangle$.

Using the above, and linearity, we obtained $U|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(U|0\rangle|0\rangle + U|1\rangle|0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$, which yields an immediate contradiction since $|\Phi^+\rangle$ is an entangled state, while $|+\rangle|+\rangle$ – the supposed output, assuming that U clones $|+\rangle$ – is a product state. □

Proof # 2. Let $|\psi_1\rangle = |\psi\rangle|0\rangle$ and $|\phi_1\rangle = |\phi\rangle|0\rangle$ be two arbitary, distinct states, and suppose that U is a unitary operator such that $U|\rho\rangle|0\rangle = |\rho\rangle|\rho\rangle$ for $\rho \in \{\psi, \phi\}$.

Consider first the inner product of $\langle\psi_1|\phi_1\rangle = (\langle\psi|\langle 0|)(|\phi\rangle|0\rangle) = (\langle\psi|\phi\rangle)(\langle 0|0\rangle) = \langle\psi|\phi\rangle$.

Alternately, the inner product $(\langle\psi|\langle\psi|)(|\phi\rangle|\phi\rangle) = (\langle\psi|\phi\rangle)^2$.

Since a unitary operator preserves inner product, this implies that $(\langle\psi|\phi\rangle)^2 = \langle\psi|\phi\rangle$, which implies that $\langle\psi|\phi\rangle \in \{0, 1\}$. This implies that U can clone any *fixed orthonormal basis*, but not arbitrary non-orthogonal states.

Note: Specifically, this means that you could build a unitary "machine" to clone $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$, but not one that can clone $\{|0\rangle, |+\rangle\}$. This is exactly why Eve is unable to implement a "copy first, measure later" attack in the BB84 protocol. □

Now that we have been introduced to some of the key tenets of quantum computing and information (superposition, interference, measurement, entanglement), let's see two applications that demonstrate their theoretical power over classical methods.

Deutsch Algorithm

The Deutsch algorithm is a foundational quantum algorithm first devised to better understand the type of problems that could be solved faster on a quantum computer. A few key things to keep in mind about the algorithm:

1. It is an algorithm conceived and analysed within the **query model of computation** – roughly, what this means is we're given a quantum black box (called an oracle) that computes some function.

2. The algorithm can query the black box with any (valid) inputs, observe the respective outputs, but – as the term black box indicates – is **not** allowed to peek inside.

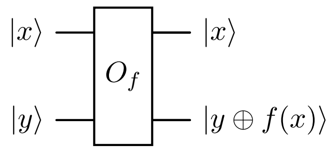
3. The objective is to **learn** the function with as few queries as possible. Indeed, within the query model, number of queries is the only measure of efficiency.

4. The black box implements a function $f : \{0, 1\} \to \{0, 1\}$, i.e. takes 0 or 1 as input, spits out a 0 or 1 as output.

5. Note that any function of this type is either **constant** (i.e. all input strings output 0 or all of them output 1) or **balanced** (exactly half of the input strings output 1). 5. The Deutsch algorithm tries to learn whether the black box f is balanced or constant.

Question: How many queries would one need (in the worst case)? (**Two.**)

Before we see how the Deutsch circuit (a quantum circuit solves this problem with **one** query), let us see how we could implement the aforementioned quantum black box. The picture below is O_f, a unitary operator that realizes f.

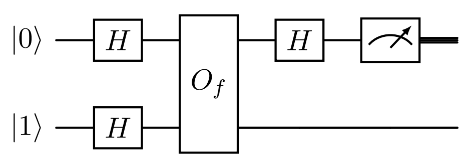


where $O_f : |x\rangle\, |y\rangle \mapsto |x\rangle\, |y \oplus f(x)\rangle$.

The inner structure of this operator is currently unimportant, but note that $|y\rangle = |0\rangle$ as the second input yields $f(x)$ in the second output wire. It is also not hard to see that for each choice of f, O_f is unitary.

Let us first explore a more "naive" approach that attempts to exploit superposition to "extract" both of the function outputs at once. Let $|x\rangle = a|0\rangle + b|1\rangle$, and let $|\psi\rangle = O_f[(a|0\rangle + b|1\rangle) \otimes |0\rangle]$. Applying O_f as defined above, we get $|\psi\rangle = a|0\rangle|f(0)\rangle + b|1\rangle|f(1)\rangle$. It appears in the above superposition, we have obtained both values of f by querying O_f exactly once. However, we can extract, via measurement, at most one of these two values.

The key here is in understanding that the problem does not require that we know the actual values of $f(0)$ and $f(1)$, merely the value of $f(0) \oplus f(1)$ – 0 if f is constant, 1 if balanced. We can extract this information in one query, using constructive/destructive interference. The complete Deutsch circuit is below.



Let us analyze each step of the circuit. We begin with $|\psi_0\rangle = |0\rangle|1\rangle$. After applying the H-gate to each qubit, we obtain $|\psi_1\rangle = |+\rangle|-\rangle$. After expanding using bilinearity, and applying the O_f operator, we obtain

$$|\psi_2\rangle = \frac{|0\rangle|0 \oplus f(0)\rangle - |0\rangle|1 \oplus f(0)\rangle + |1\rangle|0 \oplus f(1)\rangle - |1\rangle|1 \oplus f(1)\rangle}{2}.$$

To simplify the formula, verify (as an exercise) that for any bit b,

$$|0 \oplus b\rangle - |1 \oplus b\rangle = (-1)^b \left(|0\rangle - |1\rangle\right).$$

In the context of the algorithm, what the above shows is that the action of the oracle on the second qubit is converted into a phase $(-1)^{f(x)}$ on the first qubit – an important phenomenon known as *phase kickback*.

We can now rewrite

$$|\psi_2\rangle = \frac{(-1)^{f(0)}|0\rangle \left(|0\rangle - |1\rangle\right) + (-1)^{f(1)}|1\rangle \left(|0\rangle - |1\rangle\right)}{2}.$$

The analysis now splits into two cases.

Case 1: f is constant

In this case, $(-1)^{f(0)} = (-1)^{f(1)}$, and up to global phase (disregarding a global phase of -1, if necessary), we can write $|\psi_2\rangle = |+\rangle|-\rangle$. Now, applying an H-gate to the first qubit yields $|\psi_3\rangle = |0\rangle|-\rangle$. Measuring the first qubit now yields 0 with certainty. In other words, the amplitudes interfere constructively on $|0\rangle$, and destructively on $|1\rangle$.

Case 2: f is balanced

In this case, $(-1)^{f(0)} \neq (-1)^{f(1)}$; again, up to global phase (disregarding a global phase of -1, if necessary), we can write $|\psi_2\rangle = |-\rangle|-\rangle$. Now, applying an H-gate to the first qubit yields $|\psi_3\rangle = |1\rangle|-\rangle$. Measuring the first qubit now yields 1 with certainty. Analogous to the previous case, the amplitudes here interfere constructively on $|1\rangle$, and destructively on $|0\rangle$.

Conceptual summary. Thus, a single query to O_f helps us determine whether f is constant or balanced. The algorithm makes crucial use of quantum interference. The oracle O_f does not reveal the values of $f(0)$ and $f(1)$ individually; instead, via phase kickback, it encodes them as relative phases in a superposition of computational paths. The final Hadamard gate on the first qubit then causes these paths to interfere: when f is constant, the amplitudes corresponding to $|1\rangle$ interfere destructively while those for $|0\rangle$ interfere constructively, whereas when f is balanced the opposite occurs. As a result, a single measurement deterministically distinguishes the two cases, illustrating how quantum algorithms can exploit interference rather than pointwise evaluation.

The CHSH Game

After observing one application of quantum interference, we now see an application of entanglement, called the CHSH game (named after Clauser, Horne, Shimony, Holt). The CHSH game falls within a class of problems called as *nonlocal games*. The set up of the game is simple. Alice (A) and Bob (B) are playing as a team against a referee (R). R poses questions x and y to A and B respectively, where $\{x, y\} \in \{0, 1\}$. In return, A and B answer with bits a and b respectively, $\{a, b\} \in \{0, 1\}$. The players win if and only if

$$a \oplus b = x \wedge y.$$

The rules of the game are that Alice and Bob may communicate (and decide upon any strategy)

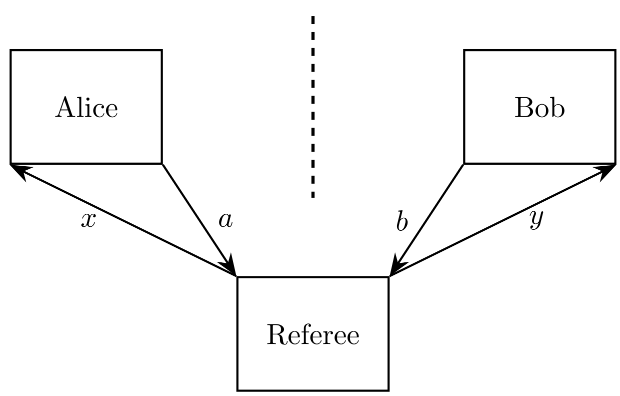


Figure 1: The CHSH game setup

before the game begins – i.e. before the referee sends them the questions – but not after the game begins.

Exercise 1. *1. Fill out the following table showing which question pairs (x, y) lead to a win for each deterministic strategy (a, b):*

(x,y) \ (a,b)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$(0,0)$				
$(0,1)$				
$(1,0)$				
$(1,1)$				

2. *For a given question pair (x, y), if Alice and Bob answer independently and uniformly at random, what is the probability of winning?*

3. *If Alice and Bob decide upon a fixed answer pair (a, b) before the game begins, what answer pairs maximize their chances of winning?*

4. *Argue that no classical strategy allows Alice and Bob to win more than 75% of the time.*

The optimal classical strategy, as evident from Problems 3 and 4, does not exceed 0.75; however, we will now study a quantum strategy that does noticeably better!

Our only assumption here will be that before the game begins, Alice and Bob share a Bell's state $|\Phi^+\rangle = \dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$. Alice and Bob will make use of the rotation gate $R_y(\theta)$. The gate can be mathematically described as below:

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

Its name signifies that it rotates a qubit $|\psi\rangle$ on the Bloch sphere by an angle of θ around the y-axis.

Exercise 2.

Prove that

$$[R_y(\theta_A) \otimes R_y(\theta_B)]|\Phi^+\rangle = \frac{\cos\left(\frac{\theta_A - \theta_B}{2}\right)(|00\rangle + |11\rangle) + \sin\left(\frac{\theta_A - \theta_B}{2}\right)(|10\rangle - |01\rangle)}{\sqrt{2}}$$

The result of Exercise 2 might seem technical, but it contains within it the key to the strategy that Alice and Bob will implement to improve their chances of winning the game. Put simply, Alice will apply a rotation $R_y(\theta_A)$ to her qubit, and Bob will apply a rotation $R_y(\theta_B)$ to his qubit, but choice of θ_A and θ_B will depend on the respective question (x and y) that they receive from the referee. The strategy is listed below, where the first value listed is for question bit 0 and the second value for question bit 1.

$$\theta_A \in \{0, \tfrac{\pi}{2}\}, \qquad \theta_B \in \{\tfrac{\pi}{4}, -\tfrac{\pi}{4}\}.$$

Note that in any scenario except when $x = y = 1$, $|\theta_A - \theta_B| = \pi/4$, while in the scenario where $x = y = 1$, $|\theta_A - \theta_B| = 3\pi/4$. Using Born's rule, we can reach the following conclusions for each (x, y) question pair. (**Exercise:** verify each of them using Exercise 2.)

1. If $x = y = 0$, then the probability that Alice and Bob measure the same bit (and thus win) is $\cos^2(\pi/8)$.
2. If $x = 0$ and $y = 1$, then the probability that Alice and Bob measure the same bit (and thus win) is $\cos^2(\pi/8)$.
3. If $x = 1$ and $y = 0$, then the probability that Alice and Bob measure the same bit (and thus win) is $\cos^2(\pi/8)$.
4. If $x = y = 1$, then the probability that Alice and Bob measure different bits (and thus win) is $\sin^2(3\pi/8) = \cos^2(\pi/8)$.

In each of the cases, the probability of winning is $\cos^2(\pi/8) \approx 0.8535$, a clear improvement over the optimal classical strategy.

Conceptual summary: The CHSH game is a canonical example of the operational power of quantum entanglement. Classically, Alice and Bob are limited to a maximum winning probability of 0.75, but quantum mechanics (entanglement in particular) allows them to exceed this bound by exploiting correlations encoded in the entangled state.